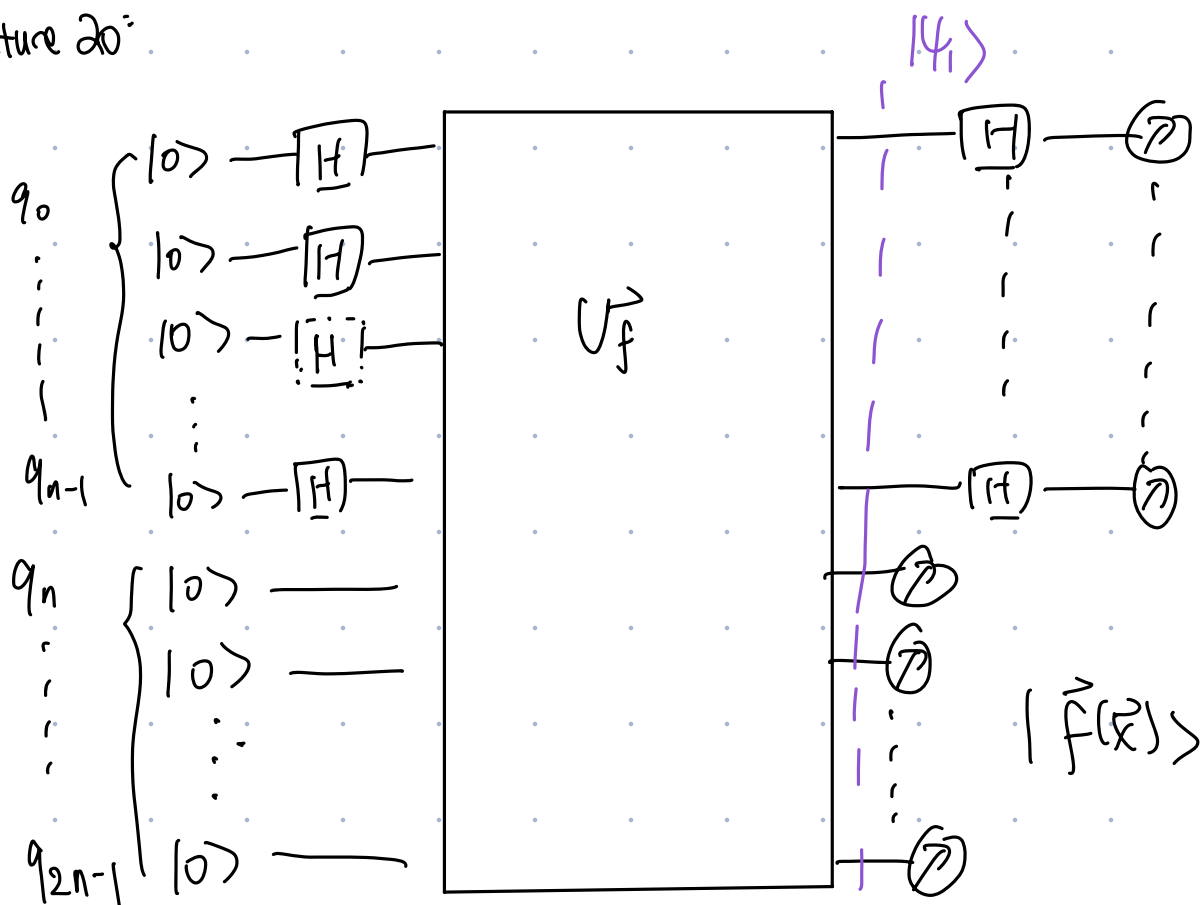


Lecture 20:



$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{x}} |f(\vec{x})\rangle \otimes |\vec{x}\rangle$$

if $\vec{b} \neq \vec{0}$

after measurement: partial collapse.

It is convenient to partition the set S of all $\vec{x}$ into $S = U(S_1, S_2)$

$$|\psi\rangle = \frac{1}{\sqrt{2^{n-1}}} \sum_{\vec{x}} |f(\vec{x})\rangle \otimes \left(\frac{1}{\sqrt{2}} |\vec{x}\rangle + |\vec{x} \oplus \vec{b}\rangle \right)$$

all unique
& orthogonal to each other.

after measurement:

$$|4_2\rangle = |f(\vec{y})\rangle \otimes \frac{1}{\sqrt{2}} (|\vec{y}\rangle + |\vec{y} \oplus \vec{b}\rangle)$$

with probability $\left| \frac{1}{\sqrt{2^{n-1}}} \right|^2 = \frac{1}{2^{n-1}}$

$\vec{y}$ any vector in S

$$P(|\vec{y}\rangle) = \frac{1}{2}$$

$$P(|\vec{y} \oplus \vec{b}\rangle) = \frac{1}{2}$$

if measure directly, we don't learn info.

$$|4^n\rangle = \frac{1}{\sqrt{2}} |f(\vec{x})\rangle \otimes H^{\otimes n} \{ |\vec{y}\rangle + |\vec{y} \oplus \vec{b}\rangle \}$$

$$= \frac{1}{\sqrt{2}} |f(\vec{x})\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{\vec{z}} (-1)^{\vec{y} \cdot \vec{z}} \{ 1 + (-1)^{\vec{b} \cdot \vec{z}} \} |\vec{z}\rangle$$

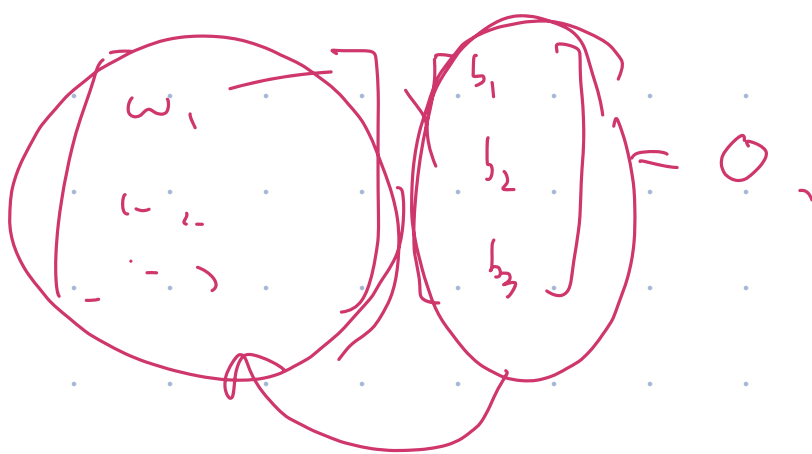
if $\vec{b} \cdot \vec{z} = 0$,

$$\{ 1 + (-1)^{\vec{b} \cdot \vec{z}} \} = 2.$$

only $|\vec{z}\rangle$ that survive are $|\vec{z}\rangle$ such that $|\vec{z}\rangle \perp \vec{b}$

$$\vec{b} \neq 0$$

$\vec{b} \cdot \vec{w} = 0 \Rightarrow$ balanced functions



$$n=2 \quad \vec{b} = (01)$$

$\vec{b} \cdot \vec{w} = 0 \rightarrow \vec{w} = \vec{0}$ is always a solution

$$\vec{w} = (10)$$

2^{n-1} solutions, one of which is $\vec{0}$.

$\hookrightarrow 2^{n-1} - 1$ nonzero solution.

if I run the algo,

half the time I get (00) , half the time I get (10) .

$\hookrightarrow$ then I can uniquely solve for $\vec{b} = (01)$.